

Signaled Bandits

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Abstract

This paper introduces the signaled bandit game, a generalization of experts' problems and bandit problems. In experts' problems the learner gets to observe the rewards of all arms independently of her selection, while in the bandit problem she only observes the reward of the selected arm. The signaled bandit game interpolates across these two regimes through the notion of signal informativeness. When the agent plays one arm, she gets to observe a noisy signal of all unplayed arms. As signals become more informative, the learner faces a reduced incentive for exploration and the game converges to the experts' setting with finite regret. Conversely, as signals become uninformative the need for dedicated exploration is restored and the game collapses to a bandit setting with asymptotic logarithmic regret. I first present a worst-case lower bound for the signaled bandit game which is smooth in signal informativeness. I then characterize an instance-dependent lower bound which features an abrupt transition from finite regret to asymptotic regret as signals become uninformative. Then, I argue that pure-exploitation algorithms cannot in general achieve sub-linear regret in the general signaled bandit, and, instead show that a simple variation of an UCB algorithm (i) preserves logarithmic or sub-logarithmic regret across the entire signaled bandit spectrum, and (ii) achieves finite regret for mildly informative signals—an impossible task for textbook UCB. Finally, I argue that signaled bandits can help us model exploration-exploitation settings in the presence of auxiliary free low quality information, including applications to portfolio selection, market entry, treatment effects and fraud detection.

1 Introduction

There exists two canonical problems in online learning: the experts' problem and the bandit problem, both with a myriad of applications to finance, marketing, clinical research, experimental design in economics and others.¹ In the stochastic version of both games, the learner selects one arm in every period and obtains a noisy reward from that arm's distribution with unknown expectation. The goal of the learner is to maximize the expected cumulative sum of rewards. The main difference across settings is that in experts' problems the learner gets to observe the outcome of counterfactual arms. By contrast, in bandit problems learner only observes the reward of the selected arm. A simple instance of the experts' problem is the *portfolio selection* problem. Every period the learner tries to conform the best possible portfolio by mixing a set of publicly traded stocks with unknown expectation. At

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the end of the period, she collects the returns of her portfolio and observes the returns of all assets, so she can compute the returns of every counterfactual portfolio. Similarly, a standard example of the bandit problem is the *AB testing* problem in marketing. Every period, the learner decides which add to show to a customer. She then observes the behavior of the customer conditional on the add, but she can never observe the customer’s behavior for the counterfactual add.

One may imagine an intermediate regime where the learner observes the reward of the selected arm and a *flow* of noisy signals for all arms independent of her selection. More concretely, we assume that signals are weakly less informative about the arm’s expectation than rewards. This notion of signal informativeness allows for a nice interpolation across regimes. When signals are as informative as rewards, the problem reduces to the experts’ game. Conversely, as signals become uninformative, the problem collapses to a bandit game. We refer to the general problem with weakly less informative signals as the **signaled bandit** game. This game can be used to model traditional exploration-exploitation settings where the learner has access to a free flow of low quality information, including applications to Finance and Economics.

Example 1 - Portfolio Selection and Public Data. Every quarter an investor selects a portfolio of non-publicly listed companies, and collects the returns of her investment at the end of the period. Although returns for counterfactual companies are not openly disclosed, the learner can get a rough idea of their returns by looking at public company reports.

Example 2 - Market Entry and Data Frequency. Every period, a firm observes some low-frequency data on demand for different markets—for instance, coming from public information releases—and decides which one it will enter for that period. Beyond capturing demand in the selected market, market entry has an additional information value to the company as it gets access to high-frequency data on demand.

Example 3 - Treatment Allocation with Scientific Communication. Consider an experimental setting where a policymaker is trying to maximize the cumulative welfare of her population by applying one in several treatments with unknown treatment effects. For instance, governments were making active decisions on vaccine suppliers or lockdown strategies during Covid-19. Assume that the ATT of each treatment is constant across populations. When selecting a treatment over her population, the policymaker can restore on publicly released health information coming from other countries. Such information is weakly less informative than internal information.

Example 4 - Fraud Detection and Effort. Many public and private agencies reward their workers by the amount of detected fraud, including tax evasion, software breaches, legal misconduct, contracts violation, etc. Every period the fraud officer inspects a case of type $k \in [K]$. For instance, cases might be split across industry, ownership, region, etc. If there is a fraud violation of scale x_1 or more, the inspector will flag it and she will receive a bounty as compensation. All unselected cases are parsed by a machine that only detects salient cases of fraud, i.e. it flags them if the fraud scale is $x_2 > x_1$. The inspector has access to the history of cases parsed by the machine, as well as her own history.

Example 5 - Treatment Allocation with Forced Exploration. Often times, the scope of experimentation is limited by regulatory or logistic considerations. For instance, a transportation company might be forced to offer a baseline service for all K routes, but she gets extra capacity to operate

additional services on these routes at a profit. Similarly, a researcher might decide how to split her time between teaching, administrative duties and research, but she must commit at least two hours in her day to each activity.

Incentives for Exploration. Intuitively, as side-signals become more informative, the learner has a reduced incentive for dedicated exploration and can simply focus on reward maximization. We can see these pattern emerging in all five examples above. For example, in the portfolio selection problem, as the quality of firm’s public data increases, the learner perceives a reduced incentive to hold assets only for information purposes. Similarly, in the case of fraud detection, as $x_2 \rightarrow x_1$, the inspector has a reduced incentive to investigate cases different from those statistically flagged by the machine. Optimal algorithms in the limiting regimes also reflect this dynamic. In the experts’ problem, pure-exploitation algorithms—like Follow-the-Leader (**FtL**)—turn out to be optimal. However, they suffer from linear regret in bandit problems. Conversely, algorithms with dedicated exploration—like Upper Confidence Bound (**UCB**)—turn out to be optimal in bandit problems but they remain overly conservative in games with full-information. This pattern suggests that an optimal algorithm in the general signaled bandit game should feature automatic exploration reduction as signals become increasingly informative.

Asymptotic vs Finite Regret. The main challenge of this interpolation exercise is the qualitatively different nature of instance-dependent regret across regimes. While finite regret is possible in the experts’ problem, no algorithm can achieve sub-logarithmic regret in the bandit problem. In other words, there is a finite-to-asymptotic regret gap between these two regimes. This abrupt transition poses the existence of a tipping point along the dimension of signal informativeness below which no algorithm can obtain finite regret, and above which finite regret becomes possible. This gap also manifests in the actual derivation of lower bounds, as standard procedures deliver empty bounds in the presence of a *linear* source of information (see [Degenne et al. \(2018\)](#)). This forces us to rely on unconventional tools to characterize the difficulty of the problem along the signal informativeness dimension [Garivier et al. \(2019\)](#). On the other hand, worst-case regret is more amenable to an interpolation analysis as both regimes display nearly-tight lower bounds on regret of $O(\sqrt{N})$.

Contributions. I first derive novel worst-case lower bounds for the general signaled bandit game of $O(\sqrt{N})$ which interpolate nicely between standard lower bounds in the experts’ and bandit problems. I leverage this result to argue that the experts’ problem is indeed the easiest instance in a signaled bandit game and the bandit problem is the hardest instance. I then characterize instance dependent lower bounds, with a particular emphasis on the finite to asymptotic regret transition point. As a glimpse, if N signals are weakly less informative than a single reward, then, the game is no different from a bandit setting and no algorithm could be expected to achieve finite regret.

In terms of algorithmic contributions, I first show that pure-exploitation algorithms cannot achieve sub-linear regret in the general signaled bandit. In particular, the regret of pure exploitation algorithms is of the same order than the informational content of signals. As a result, if the learner needs $\ln(N)$ signals to obtain the informational equivalent of a single reward, then pure-exploitation algorithms will incur $O(\ln(N))$ regret. I then show that Signaled UCB (**SUCB**)—a small variation of a UCB algorithm which uses a precision-weighted average and refined bounds—combines many desirable properties. First, it displays automatic exploration reduction. As signals become more informative, effective bounds converge to 0, so the algorithm behaves as a pure-exploitation policy. By contrast, when signals become entirely uninformative, bounds converge to standard UCB bounds. Second, I

show that this algorithm achieves finite regret for mildly informative signals and achieves sublinear regret uniformly in the signaled bandit game. In fact, the transition from asymptotic to finite regret happens at a signal informational content of $O(N\Delta)$, just like the one identified in the lower-bounds, what suggest that our lower bounds are order-tight and that SUCB is instance-dependent optimal and minmax near-optimal in the signaled bandit game.

Simulation Evidence.

Applications.

Extensions.

Our results reveal that remarkably low quality information of unplayed arms can significantly reduce the need for dedicated exploration, improving substantially the speed of learning. As such, the analyst should not neglect free information. However, completely disregarding exploration can also be problematic, especially with borderline informative signals. SUCB provides robust guarantees in the entire spectrum of signaled bandits with important applications to settings with free low-quality information.

The rest of this manuscript is structured as follows. Section 2 introduces the signaled bandit game and discusses the experts' problem and the bandit problem as especial cases. Section 3 discusses the relevant literature in Learning Theory and Economics. Section 4 derives smooth worst-case lower bounds and uses them to argue that the experts' problem is the easiest signaled bandit instance, and that the bandit problem is the hardest signaled bandit instance. We further characterize instance-dependent lower bounds and the transition point from finite to asymptotic regret. Section 5 presents an unbounded lower bound for pure-exploitation algorithms in the signaled bandit game, introduces the main algorithm of the manuscript, and shows some of its properties including automatic exploration reduction, instance-dependent optimality and near minmax optimality. Section 6 confirms the theoretical predictions through a large array of simulations. Section 7 discusses applications of signaled bandits to Economics and Finance. Section 8 considers several extensions of the main model. Finally, Section 9 concludes. All proofs can be found in Appendices A to B.

2 Model

Information Design. Consider a sequential decision problem over N periods indexed by i . Define the environment $\nu = (\nu_k)_k = (\nu_{kk'})_{kk'}$ as a collection of $K \times K$ signal distributions where $\nu_{kk'}$ is a $\sigma_{kk'}$ -subgaussian random variable with $\mathbb{E}[\nu_{kk'}] = \mu_{k'}$ for all k' .² The sub-gaussian constant σ will be the main notion of informativeness in this manuscript.³ Without loss of generality assume that $\mu_1 > \mu_2 \geq \dots \geq \mu_K$, such that $\Delta_k = \mu_1 - \mu_k > 0$ where $\Delta_k \in (0, 1]$ for all k . For simplicity we restrict our environment to the symmetric variance class, i.e.

$$\sigma_{kk'}^2 = \sigma_b^2(N) \quad \forall k \neq k', \quad \sigma_{kk}^2 = \sigma_a^2 < \infty, \quad 0 < \sigma_a^2 \leq \sigma_b^2(N) \leq \infty$$

²A random variable is σ -subgaussian if and only $\mathbb{E}[\exp(\lambda X)] \leq \exp(\lambda^2 \sigma^2 / 2)$ for all $\lambda \in \mathbb{R}_+$.

³A completely parallel analysis would go through with sub-exponential constants or domain restrictions. Although we assume that σ is known to the learner, an upper bound on σ suffices for the purpose of this analyses at the expense of some conservativeness.

The notation implies that signal informativeness $\sigma_b^2(N)$ is allowed to vary with N . Without this restriction, pure exploitation will always remain order optimal. By allowing $\sigma_b^2(N)$ to vary with N , an assumption similar to those in local-to-zero asymptotics and the weak-IV literature, we better represent frameworks with very poor yet informative signals. When $\sigma_b^2(N) = \infty$ for fixed N , we interpret that the learner receives a completely uninformative signal (or no signal at all) about unplayed arms.⁴ Every period, the learner gets to play an arm vector $k_i \in [K]$. When playing arm k in period i , she observes an *iid* stochastic realization of the signal vector $\nu_{k_i} = (\nu_{k_i,1}, \nu_{k_i,2}, \dots, \nu_{k_i,K})$ and accrues a reward $\nu_{k_i,k_i,i}$.

Definition 2.1. *Signaled Bandit Environment Class.* Let $\mathcal{E}$ be the collection of environments ν with

1. *finite mean:* $\mathbb{E}[\nu_{kk}] = \mu_k < \infty$, and *bounded gaps* $\Delta_k \in (0, 1]$ for all k ,
2. σ_a -subgaussian rewards and $\sigma_b(N)$ -subgaussian signals,
3. *finite reward informativeness* $\sigma_a^2 < \infty$ and *weakly less informative signals* $\sigma_a^2 \leq \sigma_b^2(N) \leq \infty$.

Definition 2.2. Δ *Sub-Environment Class.* For any environment class $\mathcal{E}$, we write $\mathcal{E}(\Delta) \subset \mathcal{E}$ for the Δ instance-dependent class where $\Delta_k \geq \Delta > 0$ for all $k \neq 1$.

Interpretation of Information Design. As we can see, the interpolation mechanism in this generalization of online learning games is the notion of signal informativeness. We find this mechanism more empirically grounded than alternative interpolation devices which usually rely on observing signals of side arms with the same informational content than rewards. Additionally, we find signal informativeness as more theoretically grounded than its competitors, as this concept presents a long tradition in the fields of microeconomic theory and information economics.⁵ See the literature review in Section ?? for further discussion.

The Problem of the Learner. Define the history $H_i = ((k_1, \nu_{k_1,1}), (k_2, \nu_{k_2,2}), \dots, (k_{i-1}, \nu_{k_{i-1},i-1}))$ as the collection of selected arms and signal realizations up to period i . Following our *iid* assumption, it follows that signal and reward realizations are independent of the history. Define a policy $\pi : H_i \rightarrow [K]$ as an adapted mapping from the history to the arm space $[K] = \{1, \dots, k, \dots, K\}$. Let k_i^π be the arm recommended by policy π in period i (given the history). The goal of the learner is to maximize in expectation the worst-case cumulative sum of rewards over some environment class $\mathcal{E}$, i.e.

$$\max_{\pi} \min_{\nu \in \mathcal{E}} \mathbb{E}_{\nu, \pi} \left[\sum_{i=1}^N \nu_{k_i^\pi, k_i^\pi, i} \right]$$

where the expectation is wrt to the (filtration generated by the) interaction between the history and policy. As it is customary in the learning literature, we look at the equivalent problem of regret minimization. Let $\mu^*(\nu) = \max_k \mu_k$, then

$$\min_{\pi} \max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) = \min_{\pi} \max_{\nu \in \mathcal{E}} N\mu^*(\nu) - \mathbb{E}_{\nu, \pi} \left[\sum_{i=1}^N \nu_{k_i^\pi, k_i^\pi, i} \right] \geq 0 \quad (1)$$

⁴In practice, we will see that $\sigma_b^2(N) < \infty$ for fixed N yet being equivalently uninformative to the learner.

⁵**Complete CITE.**

where $\mathcal{R}(\pi, \nu, N)$ is the regret of policy π in environment ν over N periods. A policy π is said to have regret of order $O(N^\alpha)$ in the environment class $\mathcal{E}$ iff $\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \leq C_1 \cdot N^\alpha$, for some constant $C_1 < \infty$ and $\alpha \in [0, 1]$. Similarly, an environment class $\mathcal{E}$ is said to have a lower bound of order $O(N^\beta)$ if $\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \geq C_2 \cdot N^\beta$, for some constant C_2 . Finally, a policy is said to be optimal in a class if its worst-case regret within that class matches the lower bound of the problem up to constant terms, i.e. $\alpha = \beta$. A policy is said to be near-optimal in a class if its worst-case regret within that class matches the lower bound of the problem up to logarithmic terms, i.e. $C_1 \cdot N^\alpha / (C_2 \cdot N^\beta) \leq C_3 \ln N$ for some finite constant C_3 .

Limiting and Intermediate Regimes. The signaled bandit environment is a generalization of experts' and bandit regimes.

- a) **Experts' Problems** $\mathcal{E}_{\text{exp}}$: $\mathcal{E}_{\text{exp}} = \{\nu \in \mathcal{E} : \sigma_a^2 = \sigma_b^2(N), \forall N\}$. Signals are as informative as rewards, therefore there is no trade-off between learning (exploration) and reward maximization (exploitation).
- b) **Bandit Problems** $\mathcal{E}_{\text{bandit}}$: $\mathcal{E}_{\text{bandit}} = \{\nu \in \mathcal{E} : \sigma_a^2 < M < \sigma_b^2(N) = \infty, \forall N\}$. Signals are completely uninformative while rewards remain informative. Information acquisition is tightly linked to rewards, therefore a critical trade-off emerges between learning and reward maximization.
- c) **Additional Definitions**: $\mathcal{E}_{\text{exp}}(\Delta)$ and $\mathcal{E}_{\text{bandit}}(\Delta)$ are defined as expected. $\mathcal{E}_{\text{exp}} = \mathcal{E} \setminus \mathcal{E}_{\text{exp}}$.

3 Literature Review

4 Lower Bounds in the Signaled Bandit Problem

Interpolating Lower Bounds. Our framework is smooth by construction. As the informational content of signals reduce, we should expect a smooth transition from the experts' regime to the bandit regime. One may wonder if this conceptual smoothness can be mapped into a smooth lower bound which interpolates between known results in the experts' and bandit games. The answer to this question turns out to be positive for worst-case (or minmax) lower bounds as both experts' and bandit lower bounds are of order $O(\sqrt{N})$, what eases the interpolation exercise. The task is more challenging for the case of instance-dependent bounds, where the literature has developed tight finite regret bounds in the experts' game and tight logarithmic bounds in the bandit game. In other words, there is qualitative gap between finite and asymptotic regret that hinders the smooth interpolation exercise. Yet, we leverage the smooth nature of the signaled bandit framework to (partially) characterize the regime-switching point. In particular, we show that for $\sigma_b^2(N) > 4N\Delta^2$, no algorithm can achieve finite regret in $\mathcal{E}(\Delta)$.

4.1 Minmax Lower Bounds

This section presents minimax lower bounds of order $O(\sqrt{N})$ for the signaled bandit problem with a constant that depends on the relative informational content of signals to rewards. Specifically, lower bounds are decreasing in the information content of signals, i.e. they are increasing in $\sigma_b^2(N)$. This justifies referring to the experts' problem as the easiest signaled bandit instance, and the bandit problem as the hardest. In fact, our lower bounds interpolate smoothly between known bounds in the experts' and signaled bandit games. In the K -arm problem, the lower bounds $L(\cdot)$ for the experts'

and bandit problem known are given by⁶

$$L(\text{Experts}') = C\sqrt{N\sigma_a^2} \quad L(\text{Bandit}) = C\sqrt{N(K-1)\sigma_a^2}$$

Theorem 4.1. *Lower Bound in the Signaled Bandit Problem. For any policy π ,*

$$\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) \geq \frac{1}{27} \sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2(N)}{\sigma_b^2(N) + (K-2)\sigma_a^2}}$$

Corollary 4.2. *The experts' problem is the easiest signaled bandit game and the bandit problem is the hardest signal bandit game.*

Proof of Corollary 4.2.

- a) **Experts' Problem:** For $\sigma_a^2 = \sigma_b^2(N)$, $L(\text{Signaled Bandit}) = C\sqrt{N\sigma_a^2} = L(\text{Experts})$.
- b) **Bandit Problem:** For $\sigma_a^2 < \sigma_b^2(N) = \infty$, $L(\text{Signaled Bandit}) = C\sqrt{N(K-1)\sigma_a^2} = L(\text{Bandit})$.
- c) **Signaled Bandit Problem:** For $\sigma_a^2 \leq \sigma_b^2(N) \leq \infty$ $L(\text{Signaled Bandit}) \in [L(\text{Experts}), L(\text{Bandit})]$

K Asymptotics. For any fixed N , as the number of arms goes to ∞ , the lower bounds depend only on $\sigma_b^2(N)$. This is to be expected. because for any finite σ_b^2 , the informational content of an infinite number of signals will offset the informational content of a single reward. Lower bounds reflect this principle.

4.2 Instance Dependent Lower Bounds

Consider a signaled bandit where the informational content of rewards is N times that of signals, i.e. $\sigma_b^2(N) = \sigma_a^2 N$, so the learner would need N signals about arm k to obtain the informational equivalent of a single reward from arm k . In this extreme setting, it seems obvious that the learner would need to actively play arm k if she wants to learn anything about its mean before the end of the game. In practice, this signaled bandit environment would be indistinguishable from a regular bandit, therefore, the learner cannot hope for sub-logarithmic regret. Conversely, when signals are nearly as informative as rewards, i.e. $\frac{\sigma_b^2(N)}{\sigma_a^2} = O(1)$, the learner does well in expecting finite regret by using pure-exploitation algorithms. An interesting intermediate regime emerges: what happens when $\lim_{N \rightarrow \infty} \frac{\sigma_b^2(N)}{\sigma_a^2} = \infty$, yet $\frac{\sigma_b^2(N)}{N} = o_p(1)$? Theorem 4.4 presents the main result in this section by characterizing (an upper bound on) the switching point from finite to asymptotic regret as a function of σ_b^2 .

Policy Class Restriction. As it is customary, instance-dependent bounds are only available for a sub-class of *reasonable* policies. A standard practice in bandit settings is to restrict the class of policies to algorithms that achieve at most logarithmic regret in the bandit game (Degenne et al. (2018), Lattimore and Szepesvári (2020)). There is weakly more information in the signaled bandit, compared to the bandit regime, thus the assumption is even weaker than that usually levered in the literature.

⁶See Lattimore and Szepesvári (2020) or Cesa-Bianchi and Lugosi (2006).

Assumption 4.3. *Logarithmic Policies in the Signaled Bandit Game.* We say that a policy $\pi \in \Pi_{\log}$ iff for all $N \in \mathbb{N}_+$ exist finite constants C_0, C_1 such that,

$$\max_{\nu \in \mathcal{E}(\Delta)} \mathcal{R}(\pi, \nu, N) \leq C_0 \Delta + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta}$$

Theorem 4.4. *Instance-Dependent Lower Bounds.* For all $\pi \in \Pi_{\log}$

$$\max_{\nu \in \mathcal{E}(\Delta)} \mathcal{R}(\pi, \nu, N) \geq \begin{cases} \frac{\sigma_a^2 \sigma_b^2(N)}{2\Delta(\sigma_b^2(N) - \sigma_a^2)} \left(h \left(1 \vee \frac{\sigma_b^2(N)}{4\Delta^2} \right) - \frac{1}{2} \right)_+ & \text{for } \frac{2\Delta^2 N}{h(N)} \leq \sigma_b^2(N) \leq 4N\Delta^2 \\ \frac{\sigma_a^2 \sigma_b^2(N)}{2\Delta(\sigma_b^2(N) - \sigma_a^2)} \left(h(N) - \frac{1}{2} \right)_+ & \text{for } \sigma_b^2(N) > 4N\Delta^2 \end{cases}$$

where $h(N) \sim O(\ln N)$.

Comment on Theorem 4.4. The particular definition of $h(\cdot)$, as well as the proof of Theorem 4.4 can be found in Appendix B. As anticipated, the Theorem shows that no algorithm can achieve finite regret when the informational content of signals is lower than $N\Delta^2$. Although this threshold may seem very high, note that for worst-case Δ —usually of $O(1/\sqrt{N})$ —the threshold becomes rather binding. Importantly, we note that our derivation mimics that of [Degenne et al. \(2018\)](#).⁷ In fact, the derivation of instance-dependent lower bounds in our setting features a similar challenge to the one identified in their setting: standard lower bounds become empty in the presence of a *linear* flow of action-independent information. Following their results, we overcome this problem by relying on unconventional lower-bounds as derived by [Garivier et al. \(2019\)](#). Finally, we point out that our method is tailored to the near bandit regime, as our calculations still deliver empty bounds for highly informative signals $\sigma_b^2(N) \approx \sigma_a^2$.⁸

5 Upper Bounds in the Signaled Bandit

Because our setting interpolates between the experts' and bandit regimes, one may conjecture that an algorithm which interpolates between optimal algorithms in the each of these settings is a good candidate to optimal algorithm in the general game. We show that this intuition is exactly correct. We start this Section introducing index-based algorithms, which include optimal algorithms in the limiting regimes and are amenable to an intuitive form of interpolation.

5.1 Optimal Algorithms in Limiting Regimes

In the experts' game there is no trade-off between exploration and exploitation because information is independent of the learner's actions. As a result, pure-exploitation algorithms turn out to be optimal.

Definition 5.1. *Pure-Exploitation Class.* A policy π belongs to the pure-exploitation class Π_{exploit} iff samples every arm n times (independently of N) and then selects k_i uniformly at random from $\arg \max_k \hat{\mu}_{ki}^{PE}$ for all $i \in [nK + 1, N]$, where $\hat{\mu}_{ki}^{PE} = \frac{1}{i} \sum_{s=1}^i \omega_s v_{k_s, k, s}$ for some convex weights ω_s .

⁷In [Degenne et al. \(2018\)](#), the authors derive lower bounds for a similar game to the signaled bandit where the learner observes rewards of unplayed arms with some small probability

⁸Importantly, our bounds do not explode as $\sigma_b^2 \rightarrow \sigma_a^2$. For small values of σ_b^2 the term in parenthesis becomes zero and so it does the entire bound.

In words, after a fixed-length *warm-up* exploration, pure-exploitation algorithms simply play the arm with the highest empirical mean in every period. We refer to the pure-exploitation policy with no warm-up as Follow-the-Leader (**FtL**). In the bandit regime, the learner only observes the arm of the selected arm so a critical trade-off between exploration and exploitation emerges. Upper Confidence Bound algorithms solve this trade-off through an additive bound that incentives exploration of arms with few samples. These bounds B_{ki} are selected such that $\sum_{i=1}^N \mathbb{P}(|\hat{\mu}_{ki} - \mu| \leq B_{ki})$ is conveniently small (sumable).

Definition 5.2. *Upper Confidence Bound Class.* A policy π belongs to the upper confidence bound class Π_{UCB} iff it samples every arm once and then selects k_i uniformly at random from $\arg \max_k \hat{\mu}_{ki}^B + B_{ki}$ for all $i \in [nK + 1, N]$, where $\hat{\mu}_{ki}^B = \frac{1}{N_k(i)} \sum_{s=1}^i \mathbb{1}(k_s = k) v_{k_s, k_s, s}$ and $N_k(i) = \sum_{s=1}^i \mathbb{1}(k_s = k)$

We refer to the policy $\pi \in \Pi_{\text{UCB}}$ with $B_{ki} = B_{ki}^{\text{UCB}} = \sqrt{8\sigma_a^2 \ln(i)/N_k(i)}$ as UCB, where $N_k(i) = \sum_{s=1}^i \mathbb{1}(k_s = k)$ is the number of times arm k has been played by period i .⁹ Lemma 5.3 summarizes standard results on instance-dependent and worst-case upper bounds for **FtL** and UCB.

Lemma 5.3. *Upper Bounds on Limiting Regimes.*

$$\begin{aligned} \mathcal{R}(\text{FtL}, \nu, N) &\leq \frac{16(K-1)\sigma_a^2}{\Delta} \quad \forall \nu \in \mathcal{E}_{\text{exp}}(\Delta), & \mathcal{R}(\text{UCB}, \nu, N) &\leq 4(K-1) + \frac{32 \ln(N)(K-1)\sigma_a^2}{\Delta} \quad \forall \nu \in \mathcal{E}_{\text{bandit}}(\Delta) \\ \mathcal{R}(\text{FtL}, \nu, N) &\leq 8\sqrt{N(K-1)\sigma_a^2} \quad \forall \nu \in \mathcal{E}_{\text{exp}}, & \mathcal{R}(\text{UCB}, \nu, N) &\leq 4(K-1) + 8\sqrt{2N \ln(N)(K-1)\sigma_a^2} \quad \forall \nu \in \mathcal{E}_{\text{bandit}} \end{aligned}$$

Comment on Upper Bounds. For completeness, a proof of Lemma 5.3 can be found in Appendix C. The upper bounds on these algorithms match the intuitions and lower bounds in the previous Section. **FtL** achieves finite instance-dependent regret and $O(\sqrt{N})$ worst-case regret in the experts' game. Similarly, UCB obtains logarithmic instance-dependent regret and $O(\sqrt{N \ln N})$ in the bandit game. As a result, both algorithms are optimal up to finite constants in their respective regimes (with the exception of UCB which is optimal up to logarithmic constants in the worst-case setting). Once again, the sharp gap between finite and asymptotic instance-dependent regret shows up when comparing the bandit and the experts' regime.

Shared Structure. Both algorithms are remarkably similar. Every period they both select the arm with the highest index, where the index is just a sum of an empirical average and an exploration term. The size of the exploration term is linked to the need for exploration in their respective regimes. For instance, the experts' regime features no exploration-exploitation trade-off, therefore, no learning motive. As a result, **FtL** assigns zero-weight to the exploration term, such that the arm's index reduces to its empirical average. The shared structure sets the path for a candidate to (near) optimal algorithm in the general game: Design an algorithm that selects the arm with the highest index in every period, where indexes are given by the sum of the empirical average—obtained by combining signals and rewards—and an exploration bound which decreases to 0 as signal informativeness augments.

⁹There exist more efficient algorithms in the upper confidence class compared to this particular instantiation. See for instance UCB-MOSS in Lattimore and Szepesvári (2020). Nonetheless, the gain in efficiency comes at the expense of more complicated analyses and, frequently, larger algorithmic instability. Our signaled bandit adaptation of UCB is fairly general and, we conjecture, would go through taking as reference any other algorithm in the upper confidence bound class.

5.2 A near-optimal Algorithm in the Signaled Bandit Game

Inverse Variance Weighted Sample Average. Given a set of independent observations with different information content, a reasonable estimator for the expectation of an arm is the inverse variance weighted empirical average

$$\begin{aligned}\hat{\mu}_{ki}^{\text{IVW}} &= \frac{\frac{N_k(i)}{\sigma_a^2} \frac{1}{N_k(i)} \sum_{s=1}^i \mathbb{1}(k_s = k) \nu_{k_s, k_s, s} + \frac{i - N_k(i)}{\sigma_b^2(N)} \frac{1}{i - N_k(i)} \sum_{s=1}^i \mathbb{1}(k_s \neq k) \nu_{k_s, k, s}}{\frac{N_k(i)}{\sigma_a^2} + \frac{i - N_k(i)}{\sigma_b^2(N)}} \\ &= \frac{\sigma_b^2(N) \sum_{s=1}^i \mathbb{1}(k_s = k) \nu_{k_s, k_s, s} + \sigma_a^2 \sum_{s=1}^i \mathbb{1}(k_s \neq k) \nu_{k_s, k, s}}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N) - \sigma_a^2)}\end{aligned}$$

The IVW estimator has many desirable properties. In particular, under Gauss-Markov, it is the most efficient linear estimator of the population mean. Technically, $\hat{\mu}_{ki}^{\text{IVW}}$ is not an IVW empirical average as the σ constants are sub-gaussian constants and not variances, which are unknown to the learner. However, any σ -subgaussian random variable has variance weakly less than σ^2 , so the $\hat{\mu}_{ki}^{\text{IVW}}$ estimator is a reasonable proxy for the true IVW estimator.¹⁰ $\hat{\mu}_{ki}^{\text{IVW}}$ assigns larger weights to rewards but actively features signals to construct the empirical average, especially in settings with informative signals or settings with many arms.

FtsL is Clearly Suboptimal in the Signaled Bandit Game. Before looking into more sophisticated algorithms, one may wonder if the IVW equivalent of pure-exploitation algorithms is (near) optimal in the general signaled bandit game. At the end of the day, excluding the pure bandit regime, the learner does receive non-trivial free side-information which may completely erode the need for dedicated exploration. Let Follow the Signaled Leader (**FtSL**) be the signaled bandit equivalent of **FtL**, i.e. an algorithm which selects the arm with the highest IVW empirical mean $\hat{\mu}_{ki}^{\text{IVW}}$ in every period. Based on our extensive discussion in the lower bound section, it is no surprise that **FtSL** performs poorly in the general signaled bandit regime. For finite $\sigma_b^2(N) < M$ for all N , the **FtSL** learner may expect finite regret, but when $\sigma_b^2(N) \sim O(N)$, the pure-exploitation may fail to guarantee sublinear regret. Lemma 5.4 formalizes this intuition and shows a very strong result on this regard. When $\sigma_b^2(N) \sim O(N^\alpha)$, the regret of pure-exploitation policies is lower bounded by a function of the same order as $\sigma_b^2(N)$.

Lemma 5.4. *Lower Bound on Pure-Exploitation Algorithms in the Signaled Bandit Game. Let $\nu \in \mathcal{E}(\Delta)$ be a two-arm bandit with $\nu_1 = (N(\mu_1, \sigma_a^2), N(\mu_2, \sigma_b^2(N)))$, $\nu_2 = (N(\mu_1, \sigma_b^2(N)), N(\mu_2, \sigma_a^2))$ and $\Delta = \mu_1 - \mu_2 > 0$. Let policy $\pi \in \Pi_{\text{exploit}}$, then*

$$\mathcal{R}(\nu, \pi, N) \geq C_1 \min \left\{ N\Delta, \frac{\sigma_a^2 + \sigma_b^2(N)}{\Delta} \right\}$$

In particular, when $\sigma_b^2(N) \geq C_2 \Delta f(N)$, $\mathcal{R}(\nu, \pi, N) \geq C_3 f(N)$ for some finite constants C_1, C_2, C_3 .

¹⁰Moreover, when the σ -subgaussian random variable is normally distributed, the random variable has variance σ^2 . Because the learner is essentially dealing with averages, asymptotically, $\hat{\mu}_{ki}^{\text{IVW}}$ may not be very far from the true IVW estimator.

Interpretation of Lemma 5.4. Taking a very simple environment as illustration, Lemma 5.4 argues that for finite signal informativeness σ_b^2 , pure-exploitation algorithms may achieve finite regret in the general signaled bandit game.¹¹ Nevertheless, as the informational content of signals vanishes with N , standard regret guarantees get compromised. Two immediate corollaries follow

Corollary 5.5. *Comparative Performance of FtSL in Signaled Bandit Games.*

1. *UCB dominates FtSL for all $\nu \in \mathcal{E}$ such that $\sigma_b^2(N) \gtrsim O(\sigma_a^2 \ln N)$,*
2. *FtSL incurs linear regret for all $\nu \in \mathcal{E}$ such that $\sigma_b^2(N) \gtrsim O(\Delta^2 N)$.*

Lemma 5.4 and Corollary 5.5 confirm that FtSL is clearly suboptimal in low-information regimes and disproves the belief that pure-exploitation algorithms can *survive* in signaled bandit settings with minimal information guarantees. Dedicated exploration seems necessary in the general signaled bandit class. In particular, an algorithm which completely disregards information coming from signals as UCB may worst-case dominate pure-exploitation algorithms that use the additional information like FtSL.

SUCB: A Near-Optimal Algorithm. Following the urge for dedicated exploration in the general signaled bandit, we proceed to derive high-probability bounds for the IVW estimator. To do so, we leverage the linear structure of $\hat{\mu}_{ki}^{\text{IVW}}$ and show that $\hat{\mu}_{ki}^{\text{IVW}} - \mu$ is a $\left(\frac{\sigma_a^2 \sigma_b^2(N)}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N) - \sigma_a^2)}\right)^{-1/2}$ -subgaussian random variable (see Lemma C.3). Accordingly, its high-probability bounds are given by

$$B_{ki}^{\text{IVW}} = \sqrt{\frac{8 \ln(i) \sigma_a^2 \sigma_b^2(N)}{i\sigma_a^2 + N_k(i)(\sigma_b^2(N) - \sigma_a^2)}}$$

Define the Signaled Upper Confidence Bound algorithm (SUCB) as

Algorithm 1 SUCB

Input $K, \sigma_a^2, \sigma_b^2(N), N$
for $i \in 1 : K$, **sample** $k_i = i$
for $i \in K + 1 : N$, **sample** $k_i = \arg \max_k \hat{\mu}_{ki}^{\text{IVW}} + B_{ki}^{\text{IVW}}$

SUCB is the main algorithm in this paper. We use the rest of this section to argue that it has many desirable properties. Firstly, it displays automatic exploration reduction as signals become increasingly informative, and interpolates nicely between FtL and UCB. Secondly, it is instance-dependent optimal, achieving finite regret for mildly informative signals—unlike UCB—and logarithmic regret regardless of signal informativeness—unlike FtSL. It is also minmax near-optimal up to logarithmic terms.

Definition 5.6. *Effective Exploration Bonus.* Let $B_i(\pi) \geq 0$ be the K -set of exploration bounds of an additive index policy π . Define the effective exploration bonus of policy π in period i as the maximum

¹¹Appendix C proves a finite upper bound on regret for FtSL for finite $\sigma_b^2(N)$.

pair-wise difference across individual exploration bounds, i.e.

$$B_i(\pi) = \max_{k, k' \in [K]} |B_{ki}(\pi) - B_{k'i}(\pi)|$$

Lemma 5.7. *Automatic Exploration Reduction.*

$$B_i(\text{SUCB}) \leq O\left((\sigma_b^2(N) - \sigma_a^2)\sqrt{\ln(i)/i}\right) \text{ wp } 1$$

where $B_i(\text{SUCB}) = \{B_{ki}^{\text{IVW}}\}_k$ and the probability is taken uniformly over all possible arm counts $N_k(i)$.

Lemma 5.7 highlights that effective SUCB bounds go to 0 as signals become increasingly informative. More precisely, they go to 0 linearly in $(\sigma_b^2 - \sigma_a^2)$, therefore, automatically reducing dedicated exploration as signals become informative. In addition, it is easy to see that $\hat{\mu}_{ki}^{\text{IVW}} \rightarrow \hat{\mu}_{ki}^{\text{PE}}$ as $\sigma_b^2(N) \rightarrow \sigma_a^2$ uniformly in N . It follows that SUCB effectively converges to FtL as signals become informative provided it selects the arm with the highest empirical average plus a uniform arm-independent term. Importantly, SUCB converges to FtL there where FtL is optimal: the experts' regime. A similar analysis holds as $\sigma_b^2(N) \rightarrow \infty$. In this case, $B_{ki}^{\text{IVW}} \rightarrow B_{ki}^{\text{UCB}}$, hence the effective exploration of SUCB converges to that of UCB. Furthermore, $\hat{\mu}_{ki}^{\text{IVW}} \rightarrow \hat{\mu}_{ki}^{\text{B}}$, so SUCB converges to UCB there where UCB is optimal: the bandit regime.

Theorem 5.8. *Upper Bounds of SUCB in the Signaled Bandit Game.*

$$\mathcal{R}(\text{SUCB}, \nu, N) \leq \begin{cases} (K-1) \left(4 + \min \left\{ \frac{32\sigma_a^2\sigma_b^2 \ln\left(\frac{32\sigma_b^2}{\Delta^2}\right)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C} \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ (K-1) \left(4 + \min \left\{ \frac{32\sigma_a^2\sigma_b^2 \ln(N)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C} \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32}, \end{cases} \quad \forall \nu \in \mathcal{E}(\Delta)$$

Moreover,

$$\mathcal{R}(\text{SUCB}, \nu, N) \leq 4(K-1) + \min \left\{ 8\sqrt{\frac{2N \ln(N)(K-1)\sigma_a^2\sigma_b^2(N)}{\sigma_b^2(N) - \sigma_a^2}}, O\left(\sqrt{N \ln(N)(K-1)\sigma_b^2(N)}\right) \right\} \quad \forall \nu \in \mathcal{E}$$

Theorem 5.8 summarizes the main result in this Section. The definition of $\hat{C}$ as well as the order derivation can be found in Appendix C. It shows that SUCB achieves logarithmic instance-dependent regret uniformly in the signaled bandit game, unlike pure-exploitation algorithms (see Lemma 5.4). However, it preserves some of the nice properties of pure-exploitation algorithms. In particular, it achieves finite instance-dependent regret for mildly informative signals, an impossible task for regular UCB algorithms. Additionally, observe that the transition point from finite to asymptotic regret identified in the Theorem is of the same order $O(N\Delta^2)$ than the transition point in the Lower Bound of Theorem 4.4. This suggest that our lower bounds are order-tight and that SUCB is optimal in a strong sense in the signaled bandit game.¹² The second term ensures that the bound remains meaningful as $\sigma_b^2 \rightarrow \sigma_a^2$.

These results emphasize that the learner needs not to trade between finite regret in informative

¹²There exists a window of length $O(1)$ given by $[N\Delta^2/32, 4\Delta^2 N]$ where SUCB only remains instance-dependent optimal up to a logarithmic factor.

scenarios (FtL) with uniform logarithmic guarantees (UCB). Instead, she can get robust guarantees through the automatic exploration reduction featured in SUCB. Finally, in terms of worst-case regret, the algorithm remains minmax near-optimal up to a logarithmic factor.

6 Simulations

Figure 1 plots the evolution of cumulative regret as a function of N for different σ_b^2 values. Similarly, Figure 2 plots cumulative regret in period $N = 2000$ as a function of σ_b^2 in a logarithmic scale. One can extract several takeaways from these simulation evidence, all in accordance to the analytical results in the paper. Starting from the performance of the pure-exploitation algorithms FtL and FtSL, we can see that they obtain what appears to be finite regret in the pure expert’s regime (see Lemma C.4). Yet, as $\sigma_b^2 \rightarrow \infty$ they cannot guarantee sublinear regret (see Lemma 5.4 and Corollary 5.5.ii), and significantly underperform compared to exploring algorithms (Corollary 5.5.i). This simple simulation evidence disregards the common misconception that for any finite signal informativeness σ_b^2 , pure-exploitation algorithms recover finite regret. Additionally, we point out that the IVW-powered pure-exploitation algorithm FtSL does significantly outperform the benchmark FtL. Secondly, textbook UCB guarantees logarithmic regret across all σ_b^2 domains and outperforms FtSL for $\sigma_b^2 \gtrsim O(\sigma_a^2 \ln N)$ (Lemma 5.5). Yet, for small values of σ_b^2 it remains overly conservative.

Finally, regarding SUCB we note that it achieves finite regret for small values of σ_b^2 and preserves logarithmic regret guarantees uniformly over σ_b^2 (Theorem 5.8). Additionally, in light of the simulation evidence, we conjecture that (i) SUCB uniformly outperforms textbook UCB, and (ii) that regret under SUCB is not lower bounded by the minimal regret across the interpolating policies, i.e. beyond robustness, there are real gains to convexity (see $\sigma_b^2 \in [100, 500]$ in Figure 2).

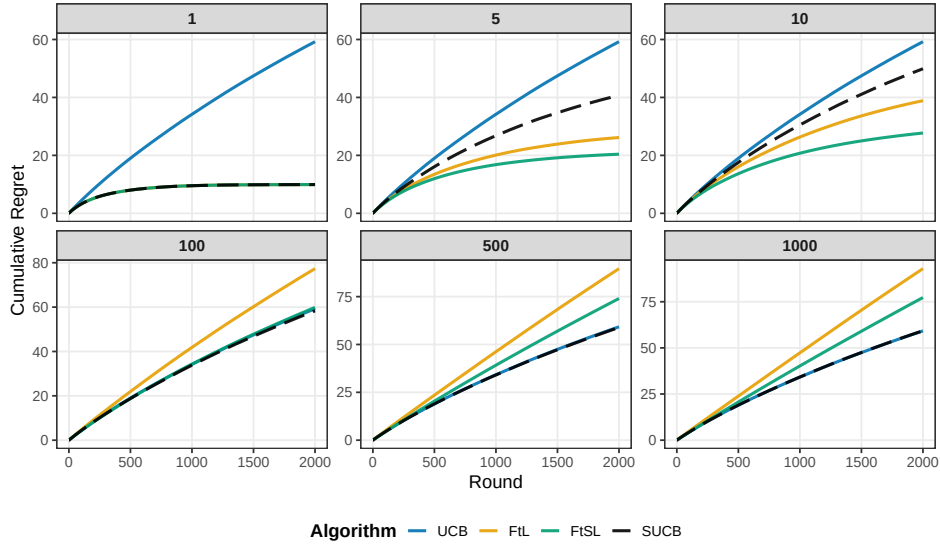


Figure 1: Cumulative Regret across σ_b^2 values (averaged over 10,000 simulations). $\mu = (0, 0.1)$, $\sigma_a^2 = 1$.

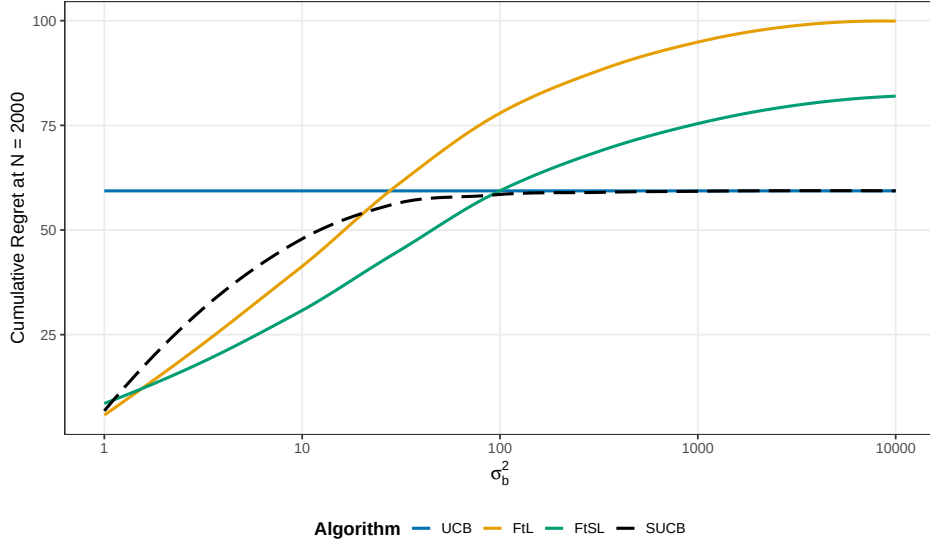


Figure 2: Cumulative Regret at period $N = 2000$ across σ_b^2 values (averaged over 1,000 simulations). $\mu = (0, 0.1)$, $\sigma_a^2 = 1$.

7 Applications

8 Extensions

9 Conclusion

This paper introduced the signaled bandit game, an online learning setting which interpolates between experts' and bandit problems through the notion of signal informativeness, parameterized by a sub-gaussian constant $\sigma^2(N)$. We argued that signal informativeness is a natural interpolation regime that synthesizes standard bandit settings with a free flow of (low quality) information with applications to finance, industrial organization, treatment effects, fraud detection and more. In terms of game difficulty, we first derived worst-case lower bounds of order $\sqrt{N}$ that interpolate smoothly between well known experts' and bandit bounds. Most notably, by allowing informativeness to scale with the horizon (similarly to local-to-zero asymptotics), we were able to derive instance-dependent bounds which feature a sharp transition between finite and logarithmic regret. We characterized the transition phase at $O(N\Delta^2)$.

In terms of algorithmic performance, we showed that pure-exploitation algorithms inherit regret of the same order as the informational content of signals, so they cannot guarantee sublinear regret across the entire signaled bandit spectrum. We then proposed a new algorithm **SUCB**, an inverse variance weighted UCB algorithm with adjusted bounds, and showed that it achieves logarithmic instance-dependent regret and $\sqrt{N \ln N}$ worst case regret in the generalized game, therefore outperforming pure-exploitation algorithms for relatively uninformative signals. More importantly, we showed that **SUCB** achieves finite instance-dependent regret for signals with informational content $\lesssim O(N\Delta^2)$, i.e. of the same order than the transition phase identified in the lower bounds. As a result, we argued that **SUCB** is order-tight in a strong sense.

Taken together, these results deliver an important message for practitioners: even low-quality information about counterfactual arms can sharply reduce the need for dedicated exploration, hence, auxiliary (unbiased) data should never be discarded. Nevertheless, exploration can still play a decisive role when the informational content of *learning by doing* is high enough. Overall, dedicated exploration yields robust guarantees at the expense of a logarithmic factor of regret. We hope the signaled bandit offers a tractable and empirically grounded lens on the many decision problems in economics and finance where costly experimentation coexists with a steady flow of free, imperfect information.

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Appendix

Before diving into the derivation of the results in this manuscript, I collect without proof a set of results and definitions that will be used repeatedly in our analysis. As a notation clarification, we will write $\sigma_b^2(N)$ as σ_b^2 unless the asymptotic properties of σ_b^2 are relevant for the analysis.

A Preliminary Results

Lemma A.1. *Regret Decomposition - Lemma 4.5 in [Lattimore and Szepesvári \(2020\)](#). For any policy π , environment $\nu \in \mathcal{E}$ and finite horizon N ,*

$$\mathcal{R}(\pi, \nu, N) = \sum_{k \neq 1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \Delta_k$$

Lemma A.2. *Linear Combinations of Subgaussian Random Variables - Lemma 5.4 and Corollary 5.5. in [Lattimore and Szepesvári \(2020\)](#). Let $\mathbb{E}[X_i] = \mu$. Assume $X_i - \mu$ are independent, σ_i -subgaussian random variables. Then,*

1. $c(X_i - \mu)$ is σ -subgaussian for all $c \in \mathbb{R}$,
2. $(X_i - \mu) + (X_j - \mu)$ is $\sqrt{\sigma_i^2 + \sigma_j^2}$ -subgaussian,

Moreover, for $\sigma_i^2 = \sigma^2$ for all i , $\hat{\mu} - \mu = \sum_{i=1}^N (X_i - \mu)/\sigma$ is $\sigma/\sqrt{N}$ -subgaussian.

Lemma A.3. *Divergence Decomposition - Lemma 15.1 in [Lattimore and Szepesvári \(2020\)](#). Consider two environments $\nu, \nu' \in \mathcal{E}$ and some policy π . Let $\mathbb{P}_{kk'}$ be the probability distribution of $\nu_{kk'}$, $\mathbb{P}_k$ be the probability distribution over the vector ν_k , $\mathbb{P}_\nu$ be the probability distribution over the environment matrix ν , and $\mathbb{P}_{\nu, \pi}$ be the probability measure induced by the N -round interaction of policy π and environment ν . $\mathbb{P}'_{kk'}$, $\mathbb{P}'_k$, $\mathbb{P}'$ and $\mathbb{P}_{\nu', \pi}$ are defined as expected. Finally, write $D(\cdot, \cdot)$ for the KL-divergence between two probability measures. Then,*

$$D(\mathbb{P}_{\nu, \pi}, \mathbb{P}_{\nu', \pi}) = \sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \cdot D(\mathbb{P}_k, \mathbb{P}'_k)$$

B Proofs of Section 4

B.1 Preliminary Lemma in Proof of Theorem 4.1

We start by presenting a small extension to Lemma A.3

Lemma B.1. *Divergence Decomposition in Signaled Bandits. Given two signaled bandits $\nu, \nu' \in \mathcal{E}$, and some policy π ,*

$$D(\mathbb{P}_{\nu, \pi}, \mathbb{P}_{\nu', \pi}) = \sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \left(\sum_{k'=1}^K D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right) \quad (2)$$

Proof of Lemma B.1. Remember that $\mathbb{P}_k = (\mathbb{P}_{k1}, \mathbb{P}_{k2}, \dots, \mathbb{P}_{kK})$. Moreover, under inde-

pendence (and the properties of logarithms) $D(\mathbb{P}_k, \mathbb{P}'_k) = \sum_{k'=1}^K D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'})$. Then,

$$\begin{aligned} D(\mathbb{P}_{\nu, \pi}, \mathbb{P}_{\nu', \pi}) &= \sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \cdot D(\mathbb{P}_k, \mathbb{P}'_k) \\ &= \sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \left(\sum_{k'=1}^K D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right) \end{aligned}$$

where the first equality follows from Lemma A.3. ■

B.2 Proof of Theorem 4.1.

The proof proceeds constructively by showing that at least one of the following two signaled bandit environments will make any policy π incur in non-trivial worst-case regret. Let ν be such that

- $\nu_{11} = N(\Delta, \sigma_a^2)$, $\nu_{k1} = N(\Delta, \sigma_b^2)$ for all $k \neq 1$,
- $\nu_{kk} = N(0, \sigma_a^2)$ for all $k \neq 1$, $\nu_{kk'} = N(0, \sigma_b^2)$ for all $k' \neq k$

Then, for some policy π and fixed horizon N , let arm l be the non-optimal vector arm with the lowest expected number of plays after N rounds, i.e.

$$l = \arg \min_{k \neq 1} \mathbb{E}_{\nu, \pi}[N_k(N)]$$

By definition of l , it follows that $\mathbb{E}_{\nu, \pi}[N_l(N)] \leq \frac{N}{K-1}$. Secondly, let ν' be such that

- $\nu'_{11} = N(\Delta, \sigma_a^2)$, $\nu'_{k1} = N(\Delta, \sigma_b^2)$ for all $k \neq 1, l$,
- $\nu'_{ll} = N(2\Delta, \sigma_a^2)$, $\nu'_{kl} = N(2\Delta, \sigma_b^2)$ for all $k \neq l$,
- $\nu_{kk} = N(0, \sigma_a^2)$ for all $k \neq 1, l$, $\nu_{kk'} = N(0, \sigma_b^2)$ for all $k' \neq k$

In other words, environment ν mimics environment ν' in every arm except for the least sampled arm ν_l . While arm 1 is optimal in ν , arm l is optimal in ν' . It follows that

$$\begin{aligned}
\mathcal{R}(\pi, \nu, N) + \mathcal{R}(\pi, \nu', N) &> \mathbb{P}_{\nu, \pi}(N_1(N) \leq N/2) \cdot \Delta \frac{N}{2} + \mathbb{P}_{\nu', \pi}(N_1(N) > N/2) \cdot \Delta \frac{N}{2} \\
&> \Delta \frac{N}{4} \exp(-D(\mathbb{P}_{\nu, \pi}, \mathbb{P}_{\nu', \pi})) \\
&= \Delta \frac{N}{4} \exp\left(-\sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \left(\sum_{k'=1}^K D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'})\right)\right) \\
&= \Delta \frac{N}{4} \exp\left(-\sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] D(\mathbb{P}_{kl}, \mathbb{P}'_{kl})\right) \\
&= \Delta \frac{N}{4} \exp\left(-\mathbb{E}_{\nu, \pi}[N_l(N)] D(\mathbb{P}_{ll}, \mathbb{P}'_{ll}) - \sum_{k \neq l}^K \mathbb{E}_{\nu, \pi}[N_k(N)] D(\mathbb{P}_{kl}, \mathbb{P}'_{kl})\right) \\
&= \Delta \frac{N}{4} \exp\left(-\mathbb{E}_{\nu, \pi}[N_l(N)] \frac{2\Delta^2}{\sigma_a^2} - \sum_{k \neq l}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \frac{2\Delta^2}{\sigma_b^2}\right) \\
&= \Delta \frac{N}{4} \exp\left(-\mathbb{E}_{\nu, \pi}[N_l(N)] 2\Delta^2 \frac{\sigma_b^2 - \sigma_a^2}{\sigma_a^2 \sigma_b^2} - \sum_{k=1}^K \mathbb{E}_{\nu, \pi}[N_k(N)] \frac{2\Delta^2}{\sigma_b^2}\right) \\
&= \Delta \frac{N}{4} \exp\left(-\mathbb{E}_{\nu, \pi}[N_l(N)] 2\Delta^2 \frac{\sigma_b^2 - \sigma_a^2}{\sigma_a^2 \sigma_b^2} - N \frac{2\Delta^2}{\sigma_b^2}\right) \\
&\geq \Delta \frac{N}{4} \exp\left(-\frac{2N\Delta^2}{K-1} \frac{\sigma_b^2 - \sigma_a^2}{\sigma_a^2 \sigma_b^2} - \frac{2N\Delta^2}{\sigma_b^2}\right) \\
&= \Delta \frac{N}{4} \exp\left(-\frac{2N\Delta^2}{K-1} \frac{\sigma_b^2 + (K-2)\sigma_a^2}{\sigma_a^2 \sigma_b^2}\right)
\end{aligned}$$

where the second inequality follows from the Bretagnole-Huber inequality, the third equality follows from Lemma B.1, the fourth equality by the selection of environments, the sixth equality by the KL-divergence of normal distributions with equal variances, and the seventh inequality by definition of arm l under policy π . Finally, selecting $\Delta = \sqrt{\frac{(K-1)\sigma_a^2\sigma_b^2}{4N(\sigma_b^2 + (K-2)\sigma_a^2)}}$ leads to

$$\begin{aligned}
\max_{\nu \in \mathcal{E}} \mathcal{R}(\pi, \nu, N) &\geq \max\{\mathcal{R}(\pi, \nu, N), \mathcal{R}(\pi, \nu', N)\} \geq \frac{\mathcal{R}(\pi, \nu, N) + \mathcal{R}(\pi, \nu', N)}{2} \\
&> \frac{\exp(-1/2)}{16} \sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2 + (K-2)\sigma_a^2}} > \frac{1}{27} \sqrt{\frac{N(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2 + (K-2)\sigma_a^2}}
\end{aligned}$$

□

B.3 Proof of Theorem 4.4

For some policy π , consider environments ν, ν' as defined in Subsection B.2. Then, starting from Lemma B.1,

$$\begin{aligned}
D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) &= \sum_{k=1}^K \mathbb{E}_{\nu,\pi}[N_k(N)] \left(\sum_{k'=1}^K D(\mathbb{P}_{kk'}, \mathbb{P}'_{kk'}) \right) \\
&= \mathbb{E}_{\nu,\pi}[N_l(N)] 2\Delta^2 \frac{\sigma_b^2 - \sigma_a^2}{\sigma_a^2 \sigma_b^2} + \frac{2\Delta^2}{\sigma_b^2} N \implies \\
\mathbb{E}_{\nu,\pi}[N_l(N)] &= D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} N
\end{aligned}$$

We now invoke Equation 6 from [Garivier et al. \(2019\)](#). In this work, the authors show that for any environment pair ν, ν' , history-adapted policy π and history-measurable random variable $Z \in [0, 1]$

$$D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) \geq D_{\text{bin}}(\mathbb{E}_{\nu,\pi}[Z], \mathbb{E}_{\nu',\pi}[Z])$$

where $D_{\text{bin}}(p, q)$ is the KL-divergence for bernoulli random variables with probability p and q , respectively. Clearly, $N_1(N)/N \in [0, 1]$ is history measurable, therefore

$$\begin{aligned}
D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) &\geq D_{\text{bin}}(\mathbb{E}_{\nu,\pi}[N_1/N], \mathbb{E}_{\nu',\pi}[N_1/N]) \\
&\geq \frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N} \cdot \ln \left(\frac{1}{\frac{\mathbb{E}_{\nu',\pi}[N_1]}{N}} \right) - \ln(2)
\end{aligned}$$

where the second inequality follows from $D_{\text{bin}}(p, q) \geq p \ln(1/q) - \ln(2)$. Now, applying Lemma [A.1](#) and Assumption [4.3](#)

$$\begin{aligned}
\mathcal{R}(\pi, \nu, N) &= \sum_{k \neq 1} \mathbb{E}_{\nu,\pi}[N_k(N)] \Delta \\
&= \Delta(N - \mathbb{E}_{\nu,\pi}[N_1(N)]) \implies \\
\frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N} &= 1 - \frac{\mathcal{R}(\pi, \nu, N)}{N\Delta} \geq 1 - \frac{1}{N} \left(C_0 + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta^2} \right)
\end{aligned}$$

and, similarly,

$$\begin{aligned}
\mathcal{R}(\pi, \nu', N) &= \mathbb{E}_{\nu',\pi}[N_1(N)] \Delta + \sum_{k \neq 1, l} \mathbb{E}_{\nu',\pi}[N_k(N)] 2\Delta \\
&\geq \mathbb{E}_{\nu',\pi}[N_1(N)] \Delta \implies \\
\frac{\mathbb{E}_{\nu',\pi}[N_1(N)]}{N} &\leq \frac{\mathcal{R}(\pi, \nu', N)}{N\Delta} \leq \frac{1}{N} \left(C_0 + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta^2} \right)
\end{aligned}$$

Finally, bringing all results together

$$\begin{aligned}
\mathbb{E}_{\nu,\pi}[N_l(N)] &= D(\mathbb{P}_{\nu,\pi}, \mathbb{P}_{\nu',\pi}) \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} N \\
&\geq \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(\frac{\mathbb{E}_{\nu,\pi}[N_1(N)]}{N} \cdot \ln \left(\frac{1}{\frac{\mathbb{E}_{\nu',\pi}[N_1]}{N}} \right) - \ln(2) \right) - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} N \\
&\geq \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(\left(1 - \frac{1}{N} \left(C_0 + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta^2} \right) \right) \cdot \ln \left(\frac{1}{\frac{1}{N} \left(C_0 + C_1 \sigma_a^2 \frac{\ln(N)}{\Delta^2} \right)} \right) - \ln(2) \right) - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} N \\
&= \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} h(N) - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} N \\
&= \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h(N) - \frac{2\Delta^2}{\sigma_b^2} N \right)_+
\end{aligned}$$

where the second-to-last equality serves as definition of $h(N)$, and the last equality emphasizes that $N_l(N)$ is non-negative, therefore the bound is only meaningful for $\sigma_b^2 \geq \frac{2\Delta^2 N}{h(N)}$

Properties of $h(N)$. We first note that $h(N) \sim O(\ln N)$. We further note that $h(N)$ is increasing for $N \geq e$. To show this, we can perform a simple analysis by taking derivatives of the first and second terms in the product and observe that they are both increasing for $N \geq \exp \left(1 - \frac{C_0 \Delta^2}{C_1 \sigma_a^2} \right)$. Some additional algebra shows that for small C_2

$$h'(N) \geq \frac{1}{N} \left(1 - \frac{1}{\ln(N) + C_2} \right)$$

such that, for $N \geq e^2$, $h'(N) \geq 1/2N$. For instance, the inequality above holds for $C_2 = \frac{C_0 \Delta^2 (\sigma_b^2 - \sigma_a^2)}{2C_1 \sigma_a^2 \sigma_b^2}$. It then follows that $\frac{\partial \mathbb{E}_{\nu,\pi}[N_l(N)]}{\partial N} \geq \frac{\sigma_a^2 \sigma_b^2}{4N \Delta^2 (\sigma_b^2 - \sigma_a^2)} - \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}$, so $\mathbb{E}_{\nu,\pi}[N_l(N)]$ is increasing (at least) in $\left[e^2, \frac{\sigma_b^2}{4\Delta^2} \right]$. Finally, we note that $\mathbb{E}_{\nu,\pi}[N_l(N)]$ is non-decreasing in N , thus,

$$\mathbb{E}_{\nu,\pi}[N_l(N)] \geq \sup_{3 \leq n \leq N} \left(\frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h(n) - \frac{2\Delta^2}{\sigma_b^2} n \right)_+ \right)$$

where 3 has been chosen as lower bound to ensure that $h(n)$ is increasing. We can now distinguish two regimes: small N and large N . For small N , $e^2 \leq N \leq \frac{\sigma_b^2}{4\Delta^2}$, from the monotonicity of $\mathbb{E}_{\nu,\pi}[N_l(N)]$, the RHS is increasing in N , therefore,

$$\begin{aligned}
\mathbb{E}_{\nu,\pi}[N_l(N)] &\geq \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h(N) - \frac{2\Delta^2}{\sigma_b^2} N \right)_+ \\
&\geq \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h(N) - \frac{1}{2} \right)_+
\end{aligned}$$

For large N , $N \geq \max \left\{ 1, \frac{\sigma_b^2}{4\Delta^2} \right\}$, the lower function can be lower bounded by its value at $n = \max \left\{ 1, \frac{\sigma_b^2}{4\Delta^2} \right\}$, thus

$$\mathbb{E}_{\nu, \pi}[N_l(N)] \geq \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h\left(1 \vee \frac{\sigma_b^2}{4\Delta^2}\right) - \frac{1}{2} \right)_+$$

Finally,

$$\mathcal{R}(\pi, \nu, N) \geq \mathbb{E}_{\nu, \pi}[N_l(N)]\Delta \geq \begin{cases} \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h\left(1 \vee \frac{\sigma_b^2}{4\Delta^2}\right) - \frac{1}{2} \right)_+ & \text{for } \frac{2\Delta^2 N}{h(N)} \leq \sigma_b^2 \leq 4N\Delta^2 \\ \frac{\sigma_a^2 \sigma_b^2}{2\Delta^2(\sigma_b^2 - \sigma_a^2)} \left(h(N) - \frac{1}{2} \right)_+ & \text{for } \sigma_b^2 > 4N\Delta^2 \end{cases}$$

□

C Proofs of Section 5

C.1 Proof of Lemma 5.4

When the learner plays arm 1, she observes (the equivalent to) a random reward $Z_i^1 = \nu_{11i} - \nu_{12i}$. Similarly, when playing arm 2, she observes (the equivalent to) a random reward $Z_i^2 = \nu_{21i} - \nu_{22i}$. Crucially, note that $Z_i^1 \sim Z_i^2 \sim N(\Delta, \sigma_a^2 + \sigma_b^2)$. Therefore, after i periods $\hat{\Delta}_i = \frac{1}{i} \sum_{s=1}^i Z_s \sim N\left(\Delta, \frac{\sigma_a^2 + \sigma_b^2}{i}\right)$. Now, observe that after round nK , the learner selects arm 2 iff $\hat{\Delta}_i < 0$. Therefore, the probability of selecting arm 2 in period i is

$$\mathbb{P}(\hat{\Delta}_i \leq 0) = \Phi\left(-\Delta \sqrt{\frac{i}{\sigma_a^2 + \sigma_b^2}}\right)$$

Now, let $m_N = \frac{\sigma_a^2 + \sigma_b^2}{\Delta^2}$. Then, observe that $\Phi(-\sqrt{i/m_N})$ is decreasing in i . Therefore $\Phi(-\sqrt{N/m_N}) \leq \Phi(-\sqrt{i/m_N})$ for all $i \leq N$. It follows that

$$\begin{aligned} \mathbb{E}[N_2(N)] &= n + \sum_{i=nK+1}^N \Phi\left(-\Delta \sqrt{\frac{i}{\sigma_a^2 + \sigma_b^2}}\right) = n + \sum_{i=nK+1}^N \Phi\left(-\sqrt{i/m_N}\right) \\ &\geq c_0 \min\{N - nK, m_N\} \end{aligned}$$

where in the last equality we have used the following result: When $m_N \geq N$, then $\Phi(\cdot) \geq \Phi(-1) = c_0$ for all i , therefore $\sum_{i=nK+1}^N \Phi\left(-\sqrt{i/m_N}\right) \geq c_0(N - nK)$. When $m_N < N$ the result above holds for all $nK + 1 \leq i \leq m_N$, i.e.

$$\begin{aligned} \sum_{i=nK+1}^N \Phi\left(-\sqrt{i/m_N}\right) &= \sum_{i=nK+1}^{m_N} \Phi\left(-\sqrt{i/m_N}\right) + \sum_{i=m_N+1}^N \Phi\left(-\sqrt{i/m_N}\right) \\ &\geq c_0(m_N - (nK + 1)) + \sum_{i=m_N+1}^N \Phi\left(-\sqrt{i/m_N}\right) \\ &\geq c_0(m_N - (nK + 1)) \end{aligned}$$

Finally, from Lemma A.1,

$$\mathcal{R}(\nu, \pi, N) = \mathbb{E}[N_2(N)]\Delta \geq c_0 \min \left\{ N\Delta, \frac{\sigma_a^2 + \sigma_b^2}{\Delta} \right\}$$

The second part of the lemma follows immediately. $\square$

C.2 Proof of Lemma 5.3

For completeness, we present quick proofs for the results in Lemma 5.3. These results are well-known to the literature and part of textbook material in Online and Bandit Learning. Our main results for the signaled bandit game will mimic the structure and tools laid down in this Subsection, therefore we see value in collecting these results here. All the expectations in this Section are taken wrt to the joint measure $\nu \times \pi$, so we write $\mathbb{E}$ for $\mathbb{E}_{\nu, \pi}$ and $\mathbb{P}$ for $\mathbb{P}_{\nu, \pi}$ whenever possible to avoid clutter.

C.3 Upper Bounds on FtL in the Experts' Problem

Note that $\frac{\mu_1 + \mu_k}{2} = \mu_k + \frac{\Delta_k}{2} = \mu_1 - \frac{\Delta_k}{2}$.

$$\begin{aligned} \mathbb{E}[N_k(N)] &\leq \mathbb{E} \left[\sum_{i=1}^N \mathbb{1}(\hat{\mu}_{ki} \geq \hat{\mu}_{1i}) \right] = \sum_{i=1}^N \mathbb{P}(\hat{\mu}_{ki} \geq \hat{\mu}_{1i}) \\ &\leq \sum_{i=1}^N \left(\mathbb{P} \left(\hat{\mu}_{ki} \geq \mu_k + \frac{\Delta_k}{2} \right) + \mathbb{P} \left(\hat{\mu}_{1i} \leq \mu_1 - \frac{\Delta_k}{2} \right) \right) \\ &\leq 2 \sum_{i=1}^N \exp \left(-\frac{i\Delta_k^2}{8\sigma_a^2} \right) \leq \frac{16\sigma_a^2}{\Delta_k^2} \end{aligned}$$

where the main inequality follows from Lemma A.2, and the last step follows from (i) bounding the geometric sum $\sum_{i=1}^N \exp \left(-\frac{i\Delta_k^2}{8\sigma_a^2} \right)$ by an infinite sum, (ii) solving the infinite sum, and (iii) noting that $\exp(-a)/(1 - \exp(-a)) \leq 1/a$. Using Lemma A.1, it now follows that for all $\nu \in \mathcal{E}_{\text{exp}}(\Delta)$

$$\begin{aligned} \mathcal{R}(\text{FtL}, \nu, N) &= \sum_{k \neq 1}^K \mathbb{E}[N_k(N)]\Delta_k \leq \sum_{k \neq 1}^K \mathbb{E}[N_k(N)]\Delta_k \\ &\leq \sum_{k \neq 1}^K \frac{16\sigma_a^2}{\Delta_k} \leq \frac{16(K-1)\sigma_a^2}{\Delta} \end{aligned}$$

Moreover, for all $\nu \in \mathcal{E}_{\text{exp}}$

$$\begin{aligned} \mathcal{R}(\text{FtL}, \nu, N) &\leq \sum_{k \neq 1, \Delta_k < \Delta}^K \mathbb{E}[N_k(N)]\Delta_k + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \mathbb{E}[N_k(N)]\Delta_k \\ &\leq N\Delta + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \frac{16\sigma_a^2}{\Delta_k} \leq N\Delta + \frac{16\sigma_a^2(K-1)}{\Delta} \\ &\leq 8\sqrt{N(K-1)\sigma_a^2} \end{aligned}$$

where the last equality follows from setting $\Delta = \sqrt{16(K-1)\sigma_a^2/N}$. $\square$

C.4 Upper Bounds on UCB in the Bandit Problem

We start by introducing some preliminary lemmas.

Lemma C.1. *Conditions for UCB. Let $B_{ki} = B_{ki}^{\text{UCB}}$ be the bound of arm k in period i . By definition of UCB, every time arm $k \neq 1$ is sampled in period $i+1$, at least one of the following conditions must be true*

$$(1) \quad \hat{\mu}_{1i} + B_{1i} \leq \mu_1 \quad (2) \quad \hat{\mu}_{ki} - B_{ki} > \mu_k \quad (3) \quad 2B_{ki} > \Delta_k \quad (3)$$

Proof of Lemma C.1. Let (2) and (3) be false, then

$$\hat{\mu}_{1i} + B_{1i} \stackrel{\text{UCB}}{\leq} \hat{\mu}_{ki} + B_{ki} = \hat{\mu}_{ki} - B_{ki} + 2B_{ki} \stackrel{(2)}{\leq} \mu_k + 2B_{ki} \stackrel{(3)}{\leq} \mu_k + \Delta_k = \mu_1$$

Let (1) and (3) be false, then

$$\hat{\mu}_{ki} - B_{ki} = \hat{\mu}_{ki} + B_{ki} - 2B_{ki} \stackrel{\text{UCB}}{\geq} \hat{\mu}_{1i} + B_{1i} - 2B_{ki} \stackrel{(1)}{\geq} \mu_1 - 2B_{ki} \stackrel{(3)}{\geq} \mu_1 - \Delta_k = \mu_k$$

Let (1) and (2) be false, then

$$\Delta_k = \mu_1 - \mu_k \stackrel{(1),(2)}{<} \hat{\mu}_{1i} + B_{1i} - (\hat{\mu}_{ki} - B_{ki}) \stackrel{\text{UCB}}{\leq} \hat{\mu}_{ki} + B_{ki} - (\hat{\mu}_{ki} - B_{ki}) = 2B_{ki}$$

■

Lemma C.2. *Probability of UCB Conditions.*

$$(3) \text{ is TRUE} \iff N_k(i) < \frac{32\sigma_a^2 \ln(i)}{\Delta_k^2}, \quad \mathbb{P}((1) \text{ is TRUE} \mid (3)) \leq i^{-3}, \quad \mathbb{P}((2) \text{ is TRUE} \mid (3)) \leq i^{-3}$$

Proof of Lemma C.2. Condition (3) is immediate given the definition of B_{ki}^{UCB} . Conditions (1) and (2) follow from Lemma A.2 and a standard *skipping argument*

$$\mathbb{P}(\hat{\mu}_{1i} + B_{1i} \leq \mu_1 \mid (3)) \leq \sum_{n=1}^i \mathbb{P}(N_k(i) = n) \mathbb{P}(\hat{\mu}_1(n) + B_1(n) \leq \mu_1) \leq \sum_{n=1}^i i^{-4} \leq i^{-3}$$

where we note that Condition (3) avoids the pathological $n = 0$ and that $\hat{\mu}_k$ is $\sigma_a/\sqrt{N_k}$ -subgaussian. Condition (2) is completely symmetric. $\blacksquare$

Remark of Lemma C.2. Set $\tilde{i}_k = \left\{ i : N_k(i) \geq \frac{32\sigma_a^2 \ln(N)}{\Delta_k^2} \right\}$, then, for all $i \geq \tilde{i}_k$, Condition (3) in Lemma C.2 is violated.

Proof of Lemma 5.3. For all $\nu \in \mathcal{E}_{\text{bandit}}(\Delta)$

$$\begin{aligned}
\mathcal{R}(\text{UCB}, \nu, N) &= \sum_{k \neq 1}^K \mathbb{E}[N_k(N)] \Delta_k = \sum_{k \neq 1}^K \sum_{i=1}^N \mathbb{P}(k_i = k) \Delta_k \\
&\leq \sum_{k \neq 1}^K \left(\tilde{i}_k \Delta_k + \sum_{i=\tilde{i}_k+1}^N \mathbb{P}(k_i = k) \Delta_k \right) \leq \sum_{k \neq 1}^K \left(\tilde{i}_k \Delta_k + 2\Delta_k \sum_{i=\tilde{i}_k+1}^N i^{-3} \right) \\
&\leq \sum_{k \neq 1}^K (\tilde{i}_k \Delta_k + 4\Delta_k) = 4 \sum_{k \neq 1}^K \Delta_k + \sum_{k \neq 1}^K \frac{32\sigma_a^2 \ln(N)}{\Delta_k} \leq 4(K-1) + \frac{32(K-1)\sigma_a^2 \ln(N)}{\Delta}
\end{aligned}$$

where the first equality follows from Lemma A.3, the fourth inequality from Lemmas C.1 and C.2, the fifth inequality from noting that $\sum_i^N i^{-3} \leq 2$ and the last inequality from noting that $\Delta_k \in [\Delta, 1]$ for all k . A similar analysis holds for all $\nu \in \mathcal{E}_{\text{bandit}}$

$$\begin{aligned}
\mathcal{R}(\text{UCB}, \nu, N) &\leq N\Delta + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \mathbb{E}[N_k(N)] \Delta_k \leq N\Delta + 4 \sum_{k \neq 1, \Delta_k \geq \Delta}^K \Delta_k + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \frac{32\sigma_a^2 \ln(N)}{\Delta_k} \\
&\leq N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2 \ln(N)}{\Delta} \leq 4(K-1) + 8\sqrt{2N \ln(N)(K-1)\sigma_a^2}
\end{aligned}$$

where the last inequality follows from setting $\Delta = \sqrt{\frac{24 \ln(N)(K-1)\sigma_a^2}{N}}$. $\square$

C.5 Subgaussianity of $\hat{\mu}_{ki}^{\text{IVW}}$

Lemma C.3. *Subgaussianity of IVW estimator. Let $\hat{\mu}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} X_{k,i}$ where $X_{k,1}, \dots, X_{k,N_k}$ is a sequence of independent σ_k -subgaussian random variables. Let*

$$\hat{\mu}_k^{\text{IVW}} = \sum_{k'=1}^K w_k \hat{\mu}_k, \quad \text{where } w_k = \frac{N_k / \sigma_k^2}{\sum_{k'=1}^K N_{k'} / \sigma_{k'}^2}$$

Then, $\hat{\mu}_k^{\text{IVW}}$ is $\left(\sum_{k=1}^K \frac{N_k}{\sigma_k^2}\right)^{-1/2}$ -subgaussian. Moreover, for any $\nu \in \mathcal{E}$, $\hat{\mu}_k^{\text{IVW}}$ is $\left(\frac{\sigma_a^2 \sigma_b^2}{N\sigma_a^2 + N_k(\sigma_b^2 - \sigma_a^2)}\right)^{1/2}$ -subgaussian.

Proof of Lemma C.3. Let $\sum_{k'}^K N_{k'} / \sigma_{k'}^2 = V$. From Lemma A.2 it follows that (i) $w_k \hat{\mu}_k$ is $w_k \sigma_k / \sqrt{N_k}$ -subgaussian and (ii) $\hat{\mu}_k^{\text{IVW}}$ is $\sqrt{\sum_{k=1}^K w_k^2 \sigma_k^2 / N_k}$ -subgaussian. Simplifying we obtain

$$\sum_{k=1}^K w_k^2 \frac{\sigma_k^2}{N_k} = \sum_{k=1}^K \left(\frac{N_k / \sigma_k^2}{V} \right)^2 = \frac{1}{V^2} \sum_{k=1}^K \frac{N_k}{\sigma_k^2} = \frac{1}{V}$$

Therefore, $\hat{\mu}_k^{\text{IVW}}$ is $(\sum_{k'}^K N_{k'} / \sigma_{k'}^2)^{-1/2}$ -subgaussian. The second part of the lemma follows from replacing σ_k^2 by σ_a^2 and σ_b^2 accordingly. $\blacksquare$

C.6 Auxiliary Proof of Upper Bound for FtSL in the Signaled Bandit Problem.

Lemma C.4. *Upper Bounds on FtSL in the Signaled bandit Game.*

$$\mathcal{R}(\text{FtSL}, \nu, N) \leq \frac{16\sigma_b^2}{\Delta_k} \leq \frac{16(K-1)\sigma_b^2}{\Delta} \quad \forall \nu \in \mathcal{E}(\Delta)$$

Moreover,

$$\mathcal{R}(\text{FtSL}, \nu, N) \leq 8\sqrt{N(K-1)\sigma_b^2} \quad \forall \nu \in \mathcal{E}$$

Proof of Lemma C.4. Recall $\hat{\mu}_{ki} = \hat{\mu}_{ki}^{\text{IVW}}(N_k(i), \sum_{k' \neq k} N_{k'}(i))$. Then,

$$\begin{aligned} \mathbb{E}[N_k(N)] &\leq \mathbb{E}\left[\sum_{i=1}^N \mathbb{1}(\hat{\mu}_{ki} > \hat{\mu}_{1i})\right] = \sum_{i=1}^N \mathbb{P}(\hat{\mu}_{ki} > \hat{\mu}_{1i}) \\ &= \sum_{i=1}^N \sum_{s=0}^i \sum_{t=0}^{i-s} \mathbb{P}(N_k(i) = s, N_1(i) = t) \cdot \mathbb{P}(\hat{\mu}_{ki}(s, i-s) > \hat{\mu}_{1i}(t, i-t)) \\ &\leq \sum_{i=1}^N \sum_{s=0}^i \sum_{t=0}^{i-s} \mathbb{P}(N_k(i) = s, N_1(i) = t) \cdot \left[\mathbb{P}\left(\hat{\mu}_{ki}(s, i-s) - \mu_k > \frac{\Delta_k}{2}\right) + \mathbb{P}\left(\mu_1 - \hat{\mu}_{1i}(t, i-t) > \frac{\Delta_k}{2}\right) \right] \\ &\leq \sum_{i=1}^N \sum_{s=0}^i \sum_{t=0}^{i-s} \mathbb{P}(N_k(i) = s, N_1(i) = t) \cdot \left[\exp\left(-\frac{\Delta_k^2(s\sigma_b^2 + (i-s)\sigma_a^2)}{8\sigma_a^2\sigma_b^2}\right) + \exp\left(-\frac{\Delta_k^2(t\sigma_b^2 + (i-t)\sigma_a^2)}{8\sigma_a^2\sigma_b^2}\right) \right] \\ &\leq \sum_{i=1}^N \sum_{s=0}^i \sum_{t=0}^{i-s} \mathbb{P}(N_k(i) = s, N_1(i) = t) \cdot \left[\exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right) + \exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right) \right] \\ &= \sum_{i=1}^N 2 \exp\left(-\frac{i\Delta_k^2}{8\sigma_b^2}\right) \leq \frac{16\sigma_b^2}{\Delta_k^2} \end{aligned}$$

where the main inequality follows from Lemma C.3. Now, using Lemma A.1, for all $\nu \in \mathcal{E}(\Delta)$

$$\mathcal{R}(\text{FtSL}, \nu, N) = \sum_{k \neq 1}^K \mathbb{E}[N_k(N)] \Delta_k \leq \sum_{k \neq 1}^K \frac{16\sigma_b^2}{\Delta_k} \leq \frac{16(K-1)\sigma_b^2}{\Delta}$$

Moreover, for all $\nu \in \mathcal{E}$

$$\begin{aligned} \mathcal{R}(\text{FtSL}, \nu, N) &= \sum_{k \neq 1, \Delta_k < \Delta}^K \mathbb{E}[N_k(N)] \Delta_k + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \mathbb{E}[N_k(N)] \Delta_k \\ &\leq N\Delta + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \frac{16\sigma_b^2}{\Delta_k} \leq N\Delta + \frac{16\sigma_b^2(K-1)}{\Delta} \\ &\leq 8\sqrt{N(K-1)\sigma_b^2} \end{aligned}$$

where the last equality follows from setting $\Delta = \sqrt{16(K-1)\sigma_b^2/N}$. $\square$

C.7 Proof of Lemma 5.7.

Note that

$$B_i(\pi) = \max_{k, k' \in [K]} |B_{ki}(\pi) - B_{k'i}(\pi)| = \max_{k \in [K]} B_{ki}(\pi) - \min_{k' \in [K]} B_{k'i}(\pi)$$

Because B_{ki} is decreasing in $N_k(i)$, $\arg \max_{k \in [K]} B_{ki}(\pi) = \arg \min_k N_k(i)$ and $\arg \min_{k' \in [K]} B_{k'i}(\pi) = \arg \max_{k'} N_{k'}(i)$. Then,

$$\begin{aligned} B_i &= \sqrt{\frac{8 \ln(i) \sigma_a^2 \sigma_b^2}{i \sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}} - \sqrt{\frac{8 \ln(i) \sigma_a^2 \sigma_b^2}{i \sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)}} \\ &= \sqrt{8 \ln(i) \sigma_a^2 \sigma_b^2} \frac{\sqrt{i \sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)} - \sqrt{i \sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)}}{\sqrt{i \sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)} \sqrt{i \sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)}} \\ &= \sqrt{8 \ln(i) \sigma_a^2 \sigma_b^2} \frac{(\sigma_b^2 - \sigma_a^2)(N_{\max}(i) - N_{\min}(i))}{\sqrt{i \sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)} \sqrt{i \sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)} \left(\sqrt{i \sigma_a^2 + N_{\max}(i)(\sigma_b^2 - \sigma_a^2)} + \sqrt{i \sigma_a^2 + N_{\min}(i)(\sigma_b^2 - \sigma_a^2)} \right)} \\ &\leq \sqrt{8 \ln(i) \sigma_a^2 \sigma_b^2} \frac{(\sigma_b^2 - \sigma_a^2)(N_{\max} - N_{\min})}{2(i \sigma_a^2)^{3/2}} \\ &\leq (\sigma_b^2 - \sigma_a^2) \frac{\sigma_b}{\sigma_a^2} \sqrt{2 \ln(i)/i} \end{aligned}$$

provided $N_{\max} - N_{\min} \leq i$. $\square$

C.8 Proof of Theorem 5.8

SUCB operates on the same logic as UCB, therefore Lemma C.1 holds immediately. We now present a small variation to Lemma C.2 for the signaled bandit game.

Lemma C.5. *Probability of SUCB Conditions.*

$$(3) \text{ is TRUE} \iff i \sigma_a^2 + N_k(i)(\sigma_b^2 - \sigma_a^2) < \frac{32 \sigma_a^2 \sigma_b^2 \ln(i)}{\Delta_k^2}, \quad \mathbb{P}((1) \text{ is TRUE} \mid (3)) \leq i^{-3}, \quad \mathbb{P}((2) \text{ is TRUE} \mid (3)) \leq i^{-3}$$

Proof of Lemma C.5. Condition (3) is immediate given the definition of B_{ki}^{IVW} . Conditions (1) and (2) follow from $\left(\frac{i \sigma_a^2 + N_k(i)(\sigma_b^2 - \sigma_a^2)}{\sigma_a^2 \sigma_b^2} \right)^{-1/2}$ -subgaussianity of $\hat{\mu}_k^{\text{IVW}}$ (see Lemma C.3) and the same skipping trick. $\blacksquare$

Remark of Lemma C.5. We distinguish two regimes which guarantee Condition (3) is violated.

1. Let $i_k^* = \min \left\{ \frac{32 \sigma_b^2}{\Delta_k^2}, N \right\}$. Observe that for $\sigma_a^2 < \sigma_b^2 < \infty$, $F(i) = \frac{32 \sigma_a^2 \sigma_b^2 \ln(i)}{\Delta_k^2 (\sigma_b^2 - \sigma_a^2)} - i \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2}$ is concave with a unique maximum in i_k^* . Let $\tilde{i}_k = \left\{ i : N_k(i) \geq \frac{32 \sigma_a^2 \sigma_b^2 \ln(i_k^*)}{\Delta_k^2 (\sigma_b^2 - \sigma_a^2)} - i_k^* \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} \right\}$, then, for all $i \geq \tilde{i}_k$, Condition (3) is violated.

2. Let $\hat{i}_k = \frac{32\sigma_b^2}{\Delta_k^2} \left(\ln \left(\frac{32\sigma_b^2}{\Delta_k^2} \right) + \ln \ln \left(\frac{32\sigma_b^2}{\Delta_k^2} \right) + \frac{32\sigma_b^2}{\Delta_k^2} \right) = \hat{C} \frac{32\sigma_b^2}{\Delta_k^2} \ln \frac{32\sigma_b^2}{\Delta_k^2}$, it follows from Lambert's W that for $i \geq \hat{i}_k$, Condition (3) is also violated for arm k .

We now use each of these regimes to derive slightly different upper bounds. Eventually, we combine both of them to obtain a tighter bound.

Proof of Theorem 5.8. (1) $\tilde{i}_k$ regime. For all $\nu \in \mathcal{E}(\Delta)$

$$\begin{aligned}
\mathcal{R}(\text{SUCB}, \nu, N) &= \sum_{k \neq 1}^K \mathbb{E}[N_k(N)] \Delta_k = \sum_{k \neq 1}^K \sum_{i=1}^N \mathbb{P}(k_i = k) \Delta_k \\
&\leq \sum_{k \neq 1}^K \left(\tilde{i}_k \Delta_k + \sum_{i=\tilde{i}_k+1}^N \mathbb{P}(k_i = k) \Delta_k \right) \leq \sum_{k \neq 1}^K \left(\tilde{i}_k \Delta_k + 2\Delta_k \sum_{i=\tilde{i}_k+1}^N i^{-3} \right) \\
&\leq \sum_{k \neq 1}^K (\tilde{i}_k \Delta_k + 4\Delta_k) = 4 \sum_{k \neq 1}^K \Delta_k + \sum_{k \neq 1}^K \left(\frac{32\sigma_a^2 \sigma_b^2 \ln(i_k^*)}{\Delta_k (\sigma_b^2 - \sigma_a^2)} - i_k^* \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} \Delta_k \right) \\
&\leq 4(K-1) + \frac{32(K-1)\sigma_a^2 \sigma_b^2 \ln \left(\min \left\{ \frac{32\sigma_b^2}{\Delta^2}, N \right\} \right)}{\Delta(\sigma_b^2 - \sigma_a^2)}
\end{aligned}$$

Result follows,

$$\mathcal{R}(\text{SUCB}, \nu, N) \leq \begin{cases} 4(K-1) + \frac{32(K-1)\sigma_a^2 \sigma_b^2 \ln \left(\frac{32\sigma_b^2}{\Delta^2} \right)}{\Delta(\sigma_b^2 - \sigma_a^2)} & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ 4(K-1) + \frac{32(K-1)\sigma_a^2 \sigma_b^2 \ln(N)}{\Delta(\sigma_b^2 - \sigma_a^2)} & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32} \end{cases}$$

Therefore, **SUCB** achieves finite regret for sufficiently small σ_b^2 . In fact, the switching point shares the same order (up to a constant factor) as the switching point identified in the lower bound.

(2) $\hat{i}_k$ regime. For all $\nu \in \mathcal{E}(\Delta)$,

$$\begin{aligned}
\mathcal{R}(\text{SUCB}, \nu, N) &= \sum_{k \neq 1}^K \mathbb{E}[N_k(N)] \Delta_k = \sum_{k \neq 1}^K \sum_{i=1}^N \mathbb{P}(k_i = k) \Delta_k \\
&= \sum_{k \neq 1}^K \left(\hat{i}_k \Delta_k + \sum_{i=\hat{i}_k+1}^N \mathbb{P}(k_i = k) \Delta_k \right) \\
&\leq \hat{C}(K-1) \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + \sum_{k \neq 1}^K \left(2\Delta_k \sum_{i=\hat{i}_k+1}^N i^{-3} \right) \\
&\leq \hat{C}(K-1) \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + \sum_{k \neq 1}^K 4\Delta_k \\
&\leq \hat{C}(K-1) \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + 4(K-1)
\end{aligned}$$

Combining both results,

$$\mathcal{R}(\text{SUCB}, \nu, N) \leq \begin{cases} (K-1) \left(4 + \min \left\{ \frac{32\sigma_a^2\sigma_b^2 \ln\left(\frac{32\sigma_b^2}{\Delta^2}\right)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C} \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 \leq \frac{N\Delta^2}{32} \\ (K-1) \left(4 + \min \left\{ \frac{32\sigma_a^2\sigma_b^2 \ln(N)}{\Delta(\sigma_b^2 - \sigma_a^2)}, \hat{C} \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} \right\} \right) & \text{if } \sigma_b^2 > \frac{N\Delta^2}{32}, \end{cases}$$

Finally, one can extend the analysis for all $\nu \in \mathcal{E}$.

$$\begin{aligned} \mathcal{R}(\text{SUCB}, \nu, N) &\leq N\Delta + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \mathbb{E}[N_k(N)]\Delta_k \\ &\leq N\Delta + 4 \sum_{k \neq 1, \Delta_k \geq \Delta}^K \Delta_k + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \left(\frac{32\sigma_a^2\sigma_b^2 \ln(i_k^*)}{\Delta_k(\sigma_b^2 - \sigma_a^2)} - i_k^* \frac{\sigma_a^2}{\sigma_b^2 - \sigma_a^2} \Delta_k \right) \\ &\leq N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln\left(\min\left\{\frac{32\sigma_b^2}{\Delta^2}, N\right\}\right)}{\Delta(\sigma_b^2 - \sigma_a^2)} \\ &\leq N\Delta + 4(K-1) + \frac{32(K-1)\sigma_a^2\sigma_b^2 \ln(N)}{\Delta} \\ &\leq 4(K-1) + 8\sqrt{\frac{2N \ln(N)(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2 - \sigma_a^2}} \end{aligned}$$

where the last inequality follows from setting $\Delta = \sqrt{\frac{32(K-1)\sigma_a^2\sigma_b^2 \ln(N)}{N(\sigma_b^2 - \sigma_a^2)}}$. Similarly,

$$\begin{aligned} \mathcal{R}(\text{SUCB}, \nu, N) &\leq N\Delta + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \mathbb{E}[N_k(N)]\Delta_k \\ &\leq N\Delta + \hat{C}(K-1) \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + \sum_{k \neq 1, \Delta_k \geq \Delta}^K \sum_{i=\hat{i}}^N \mathbb{P}(k_i = k) \Delta_k \\ &\leq N\Delta + \hat{C}(K-1) \frac{32\sigma_b^2}{\Delta} \ln \frac{32\sigma_b^2}{\Delta^2} + 4(K-1) \\ &\leq 4(K-1) + O\left(\sqrt{32\hat{C}(K-1)\sigma_b^2 \ln\left(\frac{32\sigma_b^2 N}{32\hat{C}(K-1)\sigma_b^2}\right)}\right) \\ &= 4(K-1) + O\left(\sqrt{N \ln(N)(K-1)\sigma_b^2}\right) \end{aligned}$$

where the last equality follows from setting $\Delta = \sqrt{\frac{32\hat{C}(K-1)\sigma_b^2 \ln(N)}{N}}$. Combining both results, we obtain a worst-case bound

$$\mathcal{R}(\text{SUCB}, \nu, N) \leq 4(K-1) + \min \left\{ 8\sqrt{\frac{2N \ln(N)(K-1)\sigma_a^2\sigma_b^2}{\sigma_b^2 - \sigma_a^2}}, O\left(\sqrt{N \ln(N)(K-1)\sigma_b^2}\right) \right\}$$

□